

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

AI Software Engineer

AI Software Engineer with 3+ years delivering real-time, safety-critical perception systems on edge hardware. Deep expertise in deploying vision models using PyTorch, C++17, and ROS2, with hands-on FP16 and INT8 quantisation on Intel VPUs via OpenVINO under strict latency, power and reliability constraints. Strong background translating research into deterministic, production-ready deployments through CI, profiling, and on-site validation. Experienced collaborating across AI, robotics, and hardware teams to ship resilient edge AI systems operating 24/7 in harsh, unstructured environments.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Engineered and deployed real-time CNN-based perception pipelines on edge devices using PyTorch and OpenVINO, achieving 15% F1 improvement and deterministic 24/7 operation under harsh oil-rig conditions.
- Applied FP16 quantisation with structured optimisation on Intel VPU edge hardware, meeting strict latency, memory, and stability constraints for safety-critical inference.
- Designed and validated deterministic system behaviour through on-site field testing, performing root-cause analysis of model failures and reducing crash frequency by 67% in live video pipelines.
- Refactored legacy monolith into Kafka-driven, Dockerised microservices, improving fault isolation and increasing MTBF by 35% across edge deployments.
- Designed automated data collection, augmentation, and labeling tools to scale large image datasets, increasing dataset diversity by 40% and cutting manual labeling time by 60%.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Optimised neural networks using quantisation-aware training, FP16/INT8 fixed-point inference, structured pruning, and layer fusion, reducing model size by 50% while meeting strict memory and latency budgets.
- Deployed quantised vision models to rugged Intel VPU edge devices using OpenVINO, delivering deterministic real-time inference across RGB and IR sensor streams in harsh industrial environments.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Developed adversarial-condition training and validation suites simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Built automated labelling, dataset versioning, and performance-tracking pipelines, repurposing detection models for annotation and cutting manual labelling effort by 80%, accelerating iteration cycles.
- Collaborated with hardware, software, and QA teams to integrate AI modules into production systems, mentoring engineers on optimisation and deployment practices to improve delivery consistency.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and radiological data for the teleoperation of the Clearpath Husky UGV development platform.
- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes and interfaces integrating perception outputs into a broader robotic system, managing sensor topics, tf2 frames, and message synchronisation for downstream localisation and navigation.
- Integrated perception pipelines into ROS-based robotic workflows, validating end-to-end data flow from sensors to SLAM and navigation components in Gazebo prior to deployment.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS2 prior to real-world deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 3D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

SKILLS

- **Perception & SLAM:** 3D Feature Extraction, Convolutional Neural Networks, Data Augmentation, Eigen, LiDAR SLAM, NumPy, OpenCV, Point Cloud Library, ROS2 image_transport, SAM, Scikit-Image, U-Net, Vision Transformer (ViT), YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, Docker, DVC (Data Version Control), Git, Intel VPUs, Linux, Microsoft Azure, MLFlow, NVIDIA Jetson
- **Programming & Development:** Bash, C++17, CMake, colcon, Gazebo (Harmonic), Nav2, Pandas, PostgreSQL, Python, rclcpp, rclpy, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy)
- **Deep Learning and Optimisation:** Bayesian Optimisation, DeepSORT, MultiRocket, ONNX, OpenVINO, Optuna, PyTorch, Quantisation-Aware Training, TensorFlow, TensorRT